| COs | POs | | | | | | | | | | | | PSOs | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | **PO 1** | **PO 2** | **PO 3** | **PO 4** | **PO 5** | **PO 6** | **PO 7** | **PO 8** | **PO 9** | **PO 10** | **PO 11** | **PO 12** | **PSO-1** | **PSO-2** | **PSO-3** |
| **CO-1** | 3 | 1 | 1 | 3 | 1 | - | - | - | 3 | 3 | 2 | 1 | 3 | 3 | 1 |
| **CO-2** | 2 | 3 | 1 | 3 | 2 | - | - | - | 1 | 2 | 2 | 2 | 1 | 2 | 2 |
| **CO-3** | 1 | 3 | 3 | 2 | 2 | - | - | - | 2 | 3 | 1 | 2 | 1 | 3 | 3 |
| **CO-4** | 1 | 2 | 3 | 3 | 1 | - | - | - | 3 | 2 | 1 | 3 | 3 | 3 | 3 |
| **CO-5** | 1 | 2 | 3 | 2 | 1 | - | - | - | 2 | 1 | 1 | 1 | 3 | 2 | 2 |

## CCS370 - UI AND UX DESIGN LABORATORY

## SYLLABUS

1. Designing a Responsive layout for an societal application
2. Exploring various UI Interaction Patterns
3. Developing an interface with proper UI Style Guides
4. Developing Wireflow diagram for application using open source software
5. Exploring various open source collaborative interface Platform
6. Hands on Design Thinking Process for a new product
7. Brainstorming feature for proposed product
8. Defining the Look and Feel of the new Project
9. Create a Sample Pattern Library for that product (Mood board, Fonts, Colors based on UI principles)
10. Identify a customer problem to solve
11. Conduct end-to-end user research - User research, creating personas, Ideation process (User stories, Scenarios), Flow diagrams, Flow Mapping
12. Sketch, design with popular tool and build a prototype and perform usability testing and identify improvements.

## LIST OF EXPERIMENTS

| Sl. No. | Date | Title | Page No. | Marks | Sign |
|---|---|---|---|---|---|
| 1 | | Designing a Responsive layout for an societal application | | | |
| 2 | | Exploring various UI Interaction Patterns | | | |
| 3 | | Developing an interface with proper UI Style Guides | | | |
| 4 | | Developing Wireflow diagram for application using open source software | | | |
| 5 | | Exploring various open source collaborative interface Platform | | | |
| 6 | | Hands on Design Thinking Process for a new product | | | |
| 7 | | Brainstorming feature for proposed product | | | |
| 8 | | Defining the Look and Feel of the new Project | | | |
| 9 | | Create a Sample Pattern Library for that product (Mood board, Fonts, Colors based on UI principles) | | | |
| 10 | | Identify a customer problem to solve | | | |
| 11 | | Conduct end-to-end user research - User research, creating personas, Ideation process (User stories, Scenarios), Flow diagrams, Flow Mapping | | | |
| 12 | | Sketch, design with popular tool and build a prototype and perform usability testing andidentify improvements | | | |

| EX. NO: 1 | Designing a Responsive layout for an societal application |
|---|---|
| DATE: | |

**AIM**

The aim of designing a responsive layout for a societal application in UI and UX design is to create a user-friendly, visually appealing platform that encourages engagement, inclusivity, and efficient communication, while ensuring accessibility, performance, and security.

**PROCEDURE**

**Step – 1:** Define goals and audience.

**Step – 2:** Research user preferences.

**Step – 3:** Create wireframes.

**Step – 4:** Choose design elements.

**Step – 5:** Apply responsive design principles.

**Step – 6:** Use a grid system.

**Step – 7:** Optimize typography and media.

**Step – 8:** Design intuitive navigation.

**Step – 9:** Develop using HTML/CSS/JS.

**Step – 10:** Test on various devices.

**Step – 11:** Collect user feedback.

**Step – 12:** Iterate and launch.

**Step – 13:** Continuously evaluate and update.

**PROGRAM**

**HTML**

```
<!DOCTYPE html>

<html lang="en">

<head>
```

```
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <link rel="stylesheet" href="styles.css">
    <title>Societal App</title>
</head>
<body>
    <header>
        <h1>Welcome to Societal App</h1>
    </header>
    <nav>
        <ul>
            <li><a href="#">Home</a></li>
            <li><a href="#">Posts</a></li>
            <li><a href="#">Profile</a></li>
        </ul>
    </nav>
    <main>
        <section>
            <h2>Latest Posts</h2>
            <article>
                <h3>Post Title 1</h3>
                <p>Lorem ipsum dolor sit amet, consectetur adipiscing elit...</p>
            </article>
            <article>
                <h3>Post Title 2</h3>
```

```
            <p>Proin eget ante sed ex hendrerit laoreet...</p>
        </article>
    </section>
</main>
<footer>
    <p>&copy; 2023 Societal App</p>
</footer>
<!-- Optional JavaScript inclusion -->
<script src="script.js"></script>
</body>
</html>
```

**CSS**

```
/* Reset some default browser styles */
body, h1, h2, h3, p, ul, li, a {
    margin: 0;
    padding: 0;
    list-style: none;
    text-decoration: none;
    color: #333;
}
body {
    font-family: Arial, sans-serif;
    background-color: #f0f0f0;
}
header {
```

```
    background-color: #0078d4;
    color: white;
    text-align: center;
    padding: 20px;
}
nav {
    background-color: #333;
    color: white;
    text-align: center;
    padding: 10px;
}
nav ul {
    display: flex;
    justify-content: center;
}
nav li {
    margin: 0 20px;
}
nav a {
    color: white;
    font-weight: bold;
}
main {
    padding: 20px;
}
```

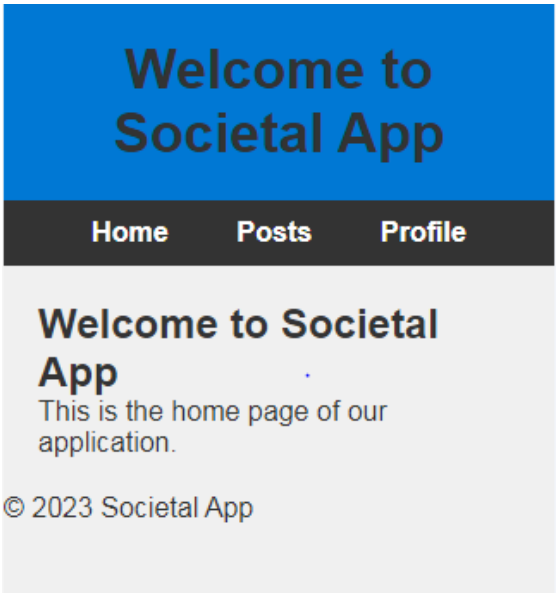
**JAVASCRIPT**

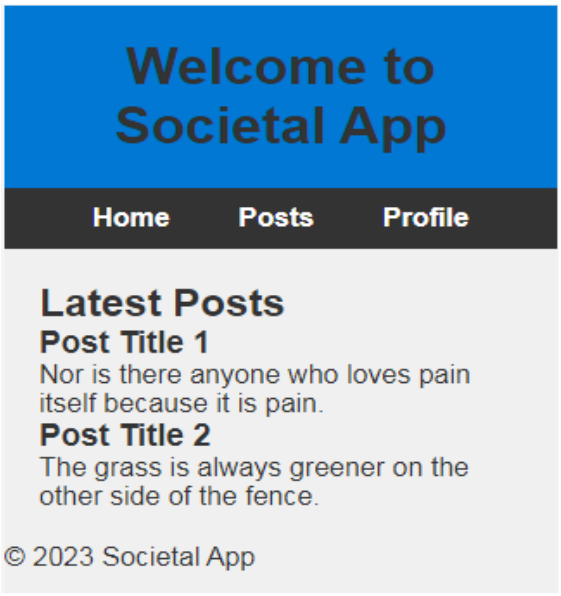
```
// Get references to the navigation links
const homeLink = document.querySelector('nav ul li:nth-child(1) a');
const postsLink = document.querySelector('nav ul li:nth-child(2) a');
const profileLink = document.querySelector('nav ul li:nth-child(3) a');

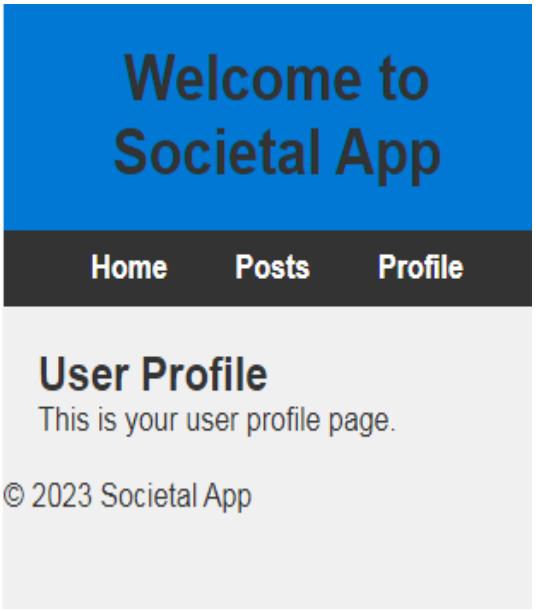

// Get references to the main content area
const mainContent = document.querySelector('main');
// Define content for different "pages"
const homeContent = `
    <h2>Welcome to Societal App</h2>
    <p>This is the home page of our application.</p>
`;
const postsContent = `
    <h2>Latest Posts</h2>
    <article>
        <h3>Post Title 1</h3>
        <p>Lorem ipsum dolor sit amet, consectetur adipiscing elit...</p>
    </article>
    <article>
        <h3>Post Title 2</h3>
        <p>Proin eget ante sed ex hendrerit laoreet...</p>
    </article>
`;
const profileContent = `
```

```
    <h2>User Profile</h2>
    <p>This is your user profile page.</p>
`;
// Add click event listeners to the navigation links
homeLink.addEventListener('click', () => {
    mainContent.innerHTML = homeContent;
});
postsLink.addEventListener('click', () => {
    mainContent.innerHTML = postsContent;
});
profileLink.addEventListener('click', () => {
    mainContent.innerHTML = profileContent;
});
// Initial content (home page)
mainContent.innerHTML = homeContent;
```

**OUTPUT**






**RESULT**

The code creates a basic responsive web page for a societal application with a header, navigation menu, main content area, and footer. Clicking on navigation links changes the content displayed in the main area, simulating page transitions. The page adapts to different screen sizes.

| EX. NO: 2 | Exploring various UI Interaction Patterns |
|---|---|
| DATE: | |

## AIM

The aim is to create an engaging and user-friendly interface by understanding and implementing various UI interaction patterns to enhance the user experience.

## PROCEDURE

**Step – 1:** Research & Objectives: Understand users and set clear goals.

**Step – 2:** Explore Patterns: Research and select suitable UI interaction patterns.

**Step – 3:** Wireframing & Prototyping: Create wireframes and interactive prototypes.

**Step – 4:** Design Elements: Define visual elements like colors and typography.

**Step – 5:** Implementation: Develop using HTML, CSS, JavaScript, or other technologies.

**Step – 6:** User Testing: Gather user feedback and test the UI interactions.

**Step – 7:** Iteration: Make improvements based on user feedback for a more engaging user experience.

## PROGRAM

## HTML

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <link rel="stylesheet" href="styles.css">
    <title>Toggle Switch Example</title>
</head>
<body>
```

```
    <label class="switch">
        <input type="checkbox" id="toggleSwitch">
        <span class="slider"></span>
    </label>
    <div id="content" class="off">
        <p>This content changes when the toggle switch is turned on.</p>
    </div>
    <script src="script.js"></script>
</body>
</html>
```

**CSS**

```
/* Basic styles for the toggle switch */
.switch {
    position: relative;
    display: inline-block;
    width: 60px;
    height: 34px;
}
.switch input {
    opacity: 0;
    width: 0;
    height: 0;
}
.slider {
    position: absolute;
```

```
    cursor: pointer;
    top: 0;
    left: 0;
    right: 0;
    bottom: 0;
    background-color: #ccc;
    -webkit-transition: .4s;
    transition: .4s;
}
.slider:before {
    position: absolute;
    content: "";
    height: 26px;
    width: 26px;
    left: 4px;
    bottom: 4px;
    background-color: white;
    -webkit-transition: .4s;
    transition: .4s;
}
input:checked + .slider {
    background-color: #2196F3;
}
input:focus + .slider {
    box-shadow: 0 0 1px #2196F3;
```

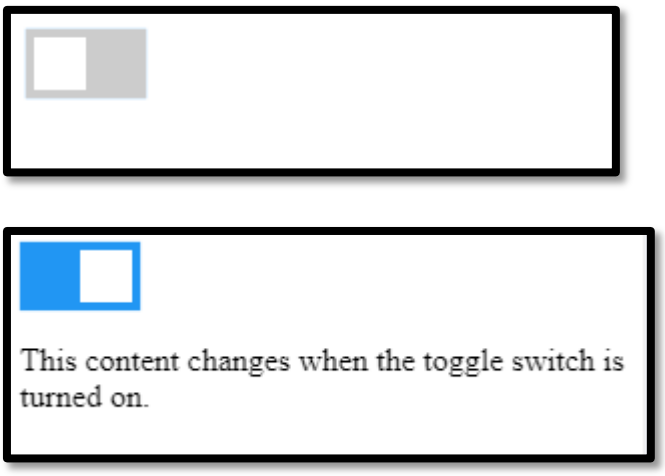
```
}
input:checked + .slider:before {
    transform: translateX(26px);
}
/* Styles for the content area */
.off {
    display: none;
}
.on {
    display: block;
}
```

**JAVASCRIPT**

```
const toggleSwitch = document.getElementById("toggleSwitch");
const content = document.getElementById("content");

toggleSwitch.addEventListener("change", function() {
    if (toggleSwitch.checked) {
        content.classList.remove("off");
        content.classList.add("on");
    } else {
        content.classList.remove("on");
        content.classList.add("off");
    }
});
```

**OUTPUT**




**RESULT**

The code creates a simple web page with a toggle switch. When the switch is turned on, the background color of the content area changes, and the content becomes visible. It illustrates the "Toggle Switch" UI interaction pattern, commonly used for enabling or disabling features.

| EX. NO: 3 | Developing an interface with proper UI Style Guides |
|---|---|
| DATE: | |

**AIM**

The aim is to create a user interface with a consistent and visually appealing design by following established UI Style Guides. This enhances usability and the overall user experience.

**PROCEDURE**

**Step – 1:** Define Goals: Clearly outline project objectives.

**Step – 2:** Select Style Guide: Choose or create a UI Style Guide.

**Step – 3:** Understand Guide: Familiarize with its design principles.

**Step – 4:** User Research: Understand user needs and preferences.

**Step – 5:** Wireframing & Prototyping: Create visuals based on the guide.

**Step – 6:** Design Elements: Apply guide's design elements (colors, typography).

**Step – 7:** UI Components: Use recommended UI components and patterns.

**Step – 8:** Coding: Write HTML, CSS, and JavaScript following the guide.

**Step – 9:** User Testing: Test with users for feedback.

**Step – 10:** Feedback & Iteration: Improve based on feedback.

**Step – 11:** Documentation: Create a reference document for consistency.

**PROGRAM**

**HTML**

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
```

```
    <link rel="stylesheet" href="styles.css">
    <title>Material Design Example</title>
</head>
<body>
    <header class="md-header">
        <h1 class="md-title">Welcome to Material Design</h1>
    </header>
    <main>
        <div class="md-card">
            <h2 class="md-card-title">Card Title</h2>
            <p class="md-card-content">This is a card with content following Material Design principles.</p>
        </div>
    </main>
    <footer class="md-footer">
        <p>&copy; 2023 Material Design Example</p>
    </footer>
</body>
</html>
```

**CSS**

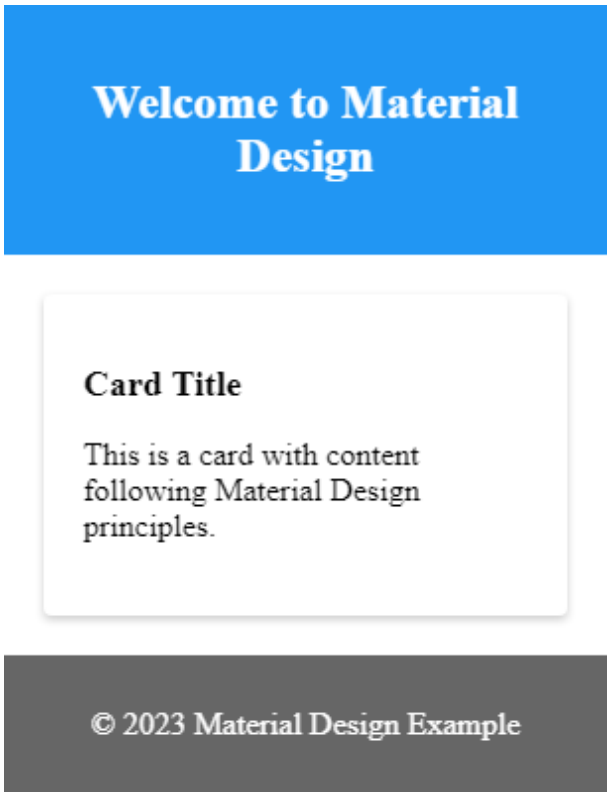
```
/* Material Design styles */
.md-header {
    background-color: #2196F3;
    color: white;
    text-align: center;
```

```
    padding: 20px;
}
.md-title {
    font-size: 24px;
}
.md-card {
    background-color: white;
    box-shadow: 0 2px 4px rgba(0, 0, 0, 0.2);
    border-radius: 4px;
    padding: 20px;
    margin: 20px;
}
.md-card-title {
    font-size: 18px;
    margin-bottom: 10px;
}
.md-card-content {
    font-size: 16px;
}
.md-footer {
    background-color: #666;
    color: white;
    text-align: center;
    padding: 10px;
}
```

**JAVASCRIPT**

```
// Get a reference to the card element
const card = document.querySelector('.md-card');
// Add a click event listener to change the card's background color
card.addEventListener('click', () => {
    // Toggle background color between white and a light Material Design blue
    if (card.style.backgroundColor === 'white' || card.style.backgroundColor === '') {
        card.style.backgroundColor = '#E3F2FD';
    } else {
        card.style.backgroundColor = 'white';
    }
});
```

**OUTPUT**



**RESULT**

The result is a visually consistent web page following Material Design principles, with a clickable card that provides a simple interactive experience. This output can serve as a foundation for more complex web applications that adhere to Material Design or similar UI Style Guides, delivering a cohesive and user-friendly user interface.

| EX. NO: 4 | |
|---|---|
| DATE: | **Developing Wireflow diagram for application using open source software** |

## AIM

The aim of developing a wireflow diagram for an application using open-source software in UI/UX design is to visually represent the user flow and interface layout, enabling efficient planning, collaboration, and communication among project stakeholders. This process helps ensure a user-centric and well-structured application design while keeping design costs in check.

## PROCEDURE

**Step – 1**: Define project scope and objectives.

**Step – 2:** Select open-source wireframing software.

**Step – 3:** Plan the user flow.

**Step – 4:** Create wireframes using the software.

**Step – 5:** Add interface elements.

**Step – 6:** Connect wireframes to show user flow.

**Step – 7:** Annotate and document the wireflow.

## PROGRAM

## HTML

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Wireflow Diagram</title>
    <style>
        body {
```

```
            font-family: Arial, sans-serif;
            margin: 20px;
        }
        h1 {
            text-align: center;
        }
        ul {
            list-style: none;
            padding: 0;
        }
        li {
            margin: 10px 0;
        }
        a {
            text-decoration: none;
            color: #0074D9;
        }
        a:hover {
            text-decoration: underline;
        }
        .page {
            margin: 20px 0;
        }
    </style>
</head>
```

```
<body>
    <h1>Wireflow Diagram</h1>
    <ul>
        <li><a href="#page1">Page 1 - Home</a></li>
        <li><a href="#page2">Page 2 - About</a></li>
        <li><a href="#page3">Page 3 - Contact</a></li>
    </ul>
    <div class="page" id="page1">
        <h2>Page 1 - Home</h2>
        <p>This is the Home page wireframe content.</p>
        <a href="#page2">Go to About</a>
    </div>
    <div class="page" id="page2">
        <h2>Page 2 - About</h2>
        <p>This is the About page wireframe content.</p>
        <a href="#page3">Go to Contact</a>
    </div>
    <div class="page" id="page3">
        <h2>Page 3 - Contact</h2>
        <p>This is the Contact page wireframe content.</p>
        <a href="#page1">Go to Home</a>
    </div>
</body>
</html>
```

**CSS**

```
/* Reset some default styles */
body, h1, h2, p {
    margin: 0;
    padding: 0;
}
body {
    font-family: Arial, sans-serif;
    margin: 20px;
    background-color: #F2F2F2;
}
h1 {
    text-align: center;
    color: #333;
    margin-top: 20px;
}
ul {
    list-style: none;
    padding: 0;
}
li {
    margin: 10px 0;
}
a {
    text-decoration: none;
    color: #0074D9;
```

```
    transition: text-decoration 0.2s;
}
a:hover {
    text-decoration: underline;
}
.page {
    background-color: #FFF;
    border: 1px solid #DDD;
    padding: 20px;
    margin: 20px 0;
    box-shadow: 0 2px 4px rgba(0, 0, 0, 0.1);
    border-radius: 5px;
}
.page h2 {
    color: #0074D9;
}
.page p {
    color: #333;
}
```

**JAVASCRIPT**

```
// JavaScript to highlight the active page link
document.addEventListener("DOMContentLoaded", function () {
    const pageLinks = document.querySelectorAll("ul li a");
    pageLinks.forEach((link) => {
        link.addEventListener("click", (event) => {
```

```
            pageLinks.forEach((l) => l.classList.remove("active"));
            event.target.classList.add("active");
        });
    });
});
```

**OUTPUT**




**RESULT**

A web page that displays a basic wireflow diagram with navigation links, wireframe content for different screens or pages, and optional interactivity if JavaScript is included. This wireflow serves as a visual representation of the application's user flow and basic interface design.

| EX. NO: 5<br>DATE: | **Exploring various open source collaborative interface Platform** |
|---|---|

**AIM**

The aim is to identify and use open-source collaborative interface platforms to enhance teamwork, streamline design processes, and create user-friendly interfaces efficiently.

**PROCEDURE**

**Step – 1:** Define project objectives.

**Step – 2:** Research open-source tools.

**Step – 3:** Evaluate features.

**Step – 4:** Select the right platform.

**Step – 5:** Onboard team members.

**Step – 6:** Collaborate on design.

**Step – 7:** Conduct user testing and gather feedback.

**PROGRAM**

**HTML**

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Collaborative Design Platform</title>
</head>
<body>
    <h1>Explore Open-Source Collaborative UI/UX Design</h1>
    <!-- Embed Figma project -->
```

```
    <div class="figma-embed">
        <!-- Replace 'YOUR_FIGMA_URL' with your Figma project URL -->
        <iframe
src="https://www.figma.com/embed?embed_host=share&url=YOUR_FIGMA_URL"
width="800" height="600" allowfullscreen></iframe>
    </div>
</body>
</html>
```

**CSS**

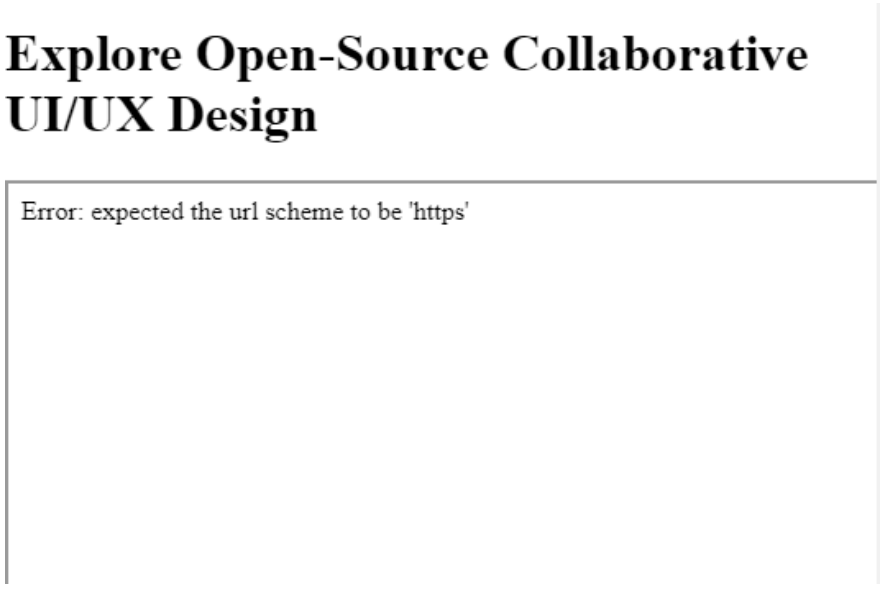
```
/* Reset some default styles */
body, h1, iframe {
    margin: 0;
    padding: 0;
}
body {
    font-family: Arial, sans-serif;
    background-color: #f2f2f2;
    margin: 20px;
}
h1 {
    text-align: center;
    padding: 20px;
}
iframe {
    display: block;
    margin: 0 auto;
```

```
    border: 1px solid #ccc;
    border-radius: 5px;
    box-shadow: 0 2px 4px rgba(0, 0, 0, 0.1);
}
```

**JAVASCRIPT**

```
// Make the iframe responsive
const iframe = document.querySelector('iframe');
function resizeIframe() {
    iframe.style.height = iframe.contentWindow.document.body.scrollHeight + 'px';
}
iframe.onload = resizeIframe;
window.addEventListener('resize', resizeIframe);
```

**OUTPUT**



**RESULT**

Exploring public source Figma URLs and design resources allows you to access pre-designed templates, find design inspiration, speed up your own design process, learn from experienced designers, and engage with the design community.

| EX. NO: 6 | Hands On Design Thinking Process For A New Product |
|---|---|
| DATE: | |

**AIM**

Create a user-friendly task management mobile app.

**PROCEDURE**

**Step - 1:** Understand user needs and problems.

**Step - 2:** Identify the problem and set goals.

**Step - 3:** Brainstorm solutions and features.

**Step - 4:** Create low-fidelity wireframes and an interactive prototype.

**Step -5:** Gather feedback on usability and design.

**Step - 6:** Write code for the app.

**Step - 7:** Ensure quality and compatibility.

**Step - 8:** Release the app to app stores.

**Step - 9:** Collect user feedback and measure success.

**Step - 10:** Make improvements based on feedback.

**PROGRAM**

**HTML**

```
<!DOCTYPE html>
<html>
<head>
    <title>Task Management App</title>
    <link rel="stylesheet" type="text/css" href="style.css">
</head>
<body>
```

```
    <header>
        <h1>Task Manager</h1>
    </header>
    <main>
        <div id="task-list">
            <!-- Tasks will be dynamically added here -->
        </div>
        <button id="add-task">Add Task</button>
    </main>
    <script src="app.js"></script>
</body>
</html>
```

**CSS**

```
body {
    font-family: Arial, sans-serif;
}
header {
    background-color: #007BFF;
    color: #fff;
    text-align: center;
    padding: 20px;
}
main {
    padding: 20px;
}
```

```
#task-list {
    margin-top: 20px;
}
#add-task {
    background-color: #007BFF;
    color: #fff;
    border: none;
    padding: 10px 20px;
    cursor: pointer;
}
```

**JAVA SCRIPT**

```
const taskList = document.getElementById('task-list');
const addTaskButton = document.getElementById('add-task');
addTaskButton.addEventListener('click', function () {
    const taskName = prompt('Enter a task:');
    if (taskName) {
        const taskItem = document.createElement('div');
        taskItem.textContent = taskName;
        taskList.appendChild(taskItem);
    }
});
```

**OUTPUT**



**RESULT**

Thu the code creates a simple task list interface for adding and displaying tasks. It's a basic prototype, not a fully functional app.

| EX. NO: 7 | Brainstroming For Proposed Product |
| --- | --- |
| DATE: | |

**AIM**

The aim is to create an intuitive, engaging, and accessible online event management platform that simplifies event organization, enhances the user experience for both event organizers and attendees, and potentially increases revenue for event organizers.

**PROCEDURE**

**Step - 1:** User registration and authentication.

**Step - 2:** Event creation and storage.

**Step - 3:** Ticketing and registration system.

**Step - 4:** Event discovery and recommendation.

**Step - 5:** Detailed event pages.

**Step - 6:** Feedback collection and ratings.

**Step - 7:** Event notifications.

**Step - 8:** Secure payment processing.

**Step - 9:** Social media integration.

**Step - 10**: Real-time chat feature.

**Step - 11:** Analytics and reporting for event organizers**.**

**Step - 12:** Ensure accessibility and inclusivity.

**PROGRAM**

**HTML**

```
<!DOCTYPE html>

<html>

<head>

    <title>Event Management Platform</title>
```

```
    <link rel="stylesheet" type="text/css" href="style.css">
</head>
<body>
    <header>
        <h1>Welcome to the Event Management Platform</h1>
    </header>
    <main>
        <section id="event-list">
            <h2>Upcoming Events</h2>
            <!-- Display a list of upcoming events here -->
            <ul>
                <li>
                    <h3>Event 1</h3>
                    <p>Date: [Event Date]</p>
                    <p>Location: [Event Location]</p>
                    <button>Register</button>
                </li>
                <!-- Repeat for other events -->
            </ul>
        </section>
        <section id="registration-form">
            <h2>Event Registration</h2>
            <form>
                <label for="name">Full Name:</label>
                <input type="text" id="name" name="name" required>
```

```
            <label for="email">Email:</label>
            <input type="email" id="email" name="email" required>
            <label for="event">Select Event:</label>
            <select id="event" name="event" required>
                <option value="event1">Event 1</option>
                <!-- Add options for other events -->
            </select>
            <button type="submit">Register</button>
        </form>
    </section>
</main>
<footer>
    <p>&copy; 2023 Event Management Platform</p>
</footer>
</body>
</html>
```

**CSS**

```
/* Reset some default browser styles */
body, h1, h2, p, ul, li, form, button, label, input, select {
    margin: 0;
    padding: 0;
}
body {
    font-family: Arial, sans-serif;
    background-color: #f0f0f0;
```

```
    margin: 0;
    padding: 0;
}
header {
    background-color: #007BFF;
    color: #fff;
    text-align: center;
    padding: 20px;
}
main {
    padding: 20px;
}
section {
    background-color: #fff;
    padding: 20px;
    margin-bottom: 20px;
    box-shadow: 0 0 5px rgba(0, 0, 0, 0.2);
}
h1, h2 {
    color: #007BFF;
}
ul {
    list-style: none;
}
li {
```

```
    margin: 10px 0;
}
button {
    background-color: #007BFF;
    color: #fff;
    border: none;
    padding: 10px 20px;
    cursor: pointer;
}
form {
    max-width: 300px;
    margin: 0 auto;
}
label, input, select {
    display: block;
    margin-bottom: 10px;
}
/* Additional styling can be added to customize your design further */
```

**JAVASCRIPT**

```
// Example front-end JavaScript for event management platform
// Simulated data for events (replace with actual data from the server)
const events = [
    { name: 'Event 1', date: '2023-01-15', location: 'Location A' },
    { name: 'Event 2', date: '2023-02-20', location: 'Location B' },
    // Add more events here
```

```
];
// Function to display events
function displayEvents() {
    const eventList = document.getElementById('event-list');
    events.forEach(event => {
        const eventItem = document.createElement('li');
        eventItem.innerHTML = `
            <h3>${event.name}</h3>
            <p>Date: ${event.date}</p>
            <p>Location: ${event.location}</p>
            <button onclick="registerForEvent('${event.name}')">Register</button>
        `;
        eventList.appendChild(eventItem);
    });
}
// Function to register for an event
function registerForEvent(eventName) {
    // Replace with code to handle event registration
    alert(`Registered for ${eventName}!`);
}
// Load events when the page loads
window.addEventListener('load', displayEvents);
```

**OUTPUT**

# Welcome to the Event Management Platform

## Upcoming Events

- **Event 1**

  Date: [Event Date]

  Location: [Event Location]

  Register

## Event Registration

Full Name: Email: Select Event: Event 1 Register

© 2023 Event Management Platform

**RESULT**

Thus the program to create an intuitive, engaging, and accessible online event management platform that simplifies event organization, enhances the user experience for both event organizers and attendees, and potentially increases revenue for event organizers is successfully executed.

| EX. NO: 8 | Defining The Look And Feel Of The New Project |
|---|---|
| DATE: | |

**AIM**

The aim for defining the look and feel of a new project in UI and UX design is to create a visually engaging and aesthetically pleasing user interface that enhances user experience, aligns with the project's goals, and resonates with the target audience.

**PROCEDURE**

**Step - 1:** Project Analysis: Understand goals and user needs.

**Step - 2:** User Persona: Create user profiles.

**Step - 3:** Competitor Analysis: Research competitors.

**Step - 4:** Wireframing and Mockups: Sketch design concepts.

**Step - 5:** Color and Typography: Select visuals.

**Step - 6:** Visual Elements: Decide icons and images.

**Step - 7**: Interactive Elements: Plan user interactions.

**Step - 8:** Accessibility: Ensure inclusivity.

**Step - 9:** User Testing: Collect feedback.

**Step - 10:** Iteration: Refine design based on feedback.

**Step - 11:** Style Guide: Document design choices.

**Step - 12:** Handoff: Share with development team.

**PROGRAM**

**HTML**

```
<!DOCTYPE html>
<html>
<head>
    <title>UI/UX Design Style Guide</title>
```

```
    <link rel="stylesheet" type="text/css" href="style.css">
</head>
<body>
    <header>
        <h1>Design Style Guide</h1>
    </header>
    <main>
        <section id="colors">
            <h2>Colors</h2>
            <div class="color-box" id="primary"></div>
            <div class="color-box" id="secondary"></div>
        </section>
        <section id="typography">
            <h2>Typography</h2>
            <p class="font-heading">Heading Font</p>
            <p class="font-body">Body Font</p>
        </section>
    </main>
    <script src="script.js"></script>
</body>
</html>
```

**CSS**

```
body {
    font-family: Arial, sans-serif;
}
```

```
header {
    background-color: #007BFF;
    color: #fff;
    text-align: center;
    padding: 20px;
}
main {
    padding: 20px;
}
h2 {
    color: #007BFF;
}
.color-box {
    width: 100px;
    height: 100px;
    margin: 10px;
    display: inline-block;
}
#primary {
    background-color: #007BFF;
}
#secondary {
    background-color: #FF6C00;
}
.font-heading {
```

```
    font-family: 'Arial', sans-serif;
    font-weight: bold;
    font-size: 24px;
}
.font-body {
    font-family: 'Arial', sans-serif;
    font-size: 16px;
}
```

**JAVASCRIPT**

```
// JavaScript for interaction (optional)
// Change the background color of the primary color box on click
document.getElementById("primary").addEventListener("click", function () {
    this.style.backgroundColor = "#FF0000";
});
// Change the background color of the secondary color box on double-click
document.getElementById("secondary").addEventListener("dblclick", function () {
    this.style.backgroundColor = "#00FF00";
});
```

**OUTPUT**

# Design Style Guide

## Colors

## Typography

Heading Font

Body Font

**RESULT**

Thus the result is a webpage with a design style guide showing color and typography elements. JavaScript allows you to interactively change the colors of the color boxes.

| EX. NO: 9<br>DATE: | **Create a Sample Pattern Library for that product (Mood board, Fonts, Colors based on UI principles)** |
|---|---|

**AIM**

The aim of this sample pattern library is to provide a comprehensive and organized resource for UI and UX design, fostering consistency, creativity, and user-centricity throughout the product development process.

**PROCEDURE**

**Step – 1:** Initiate the Design Process

**Step – 2:** Mood Board Creation

**Step – 3:** Font Selection

**Step – 4:** Color Palette Definition

**Step – 5:** UI Principles

**Step – 6:** Documentation and Collaboration

**Step – 7:** Iterative Design Process

**Step – 8:** Implementation in Design Process

**Step – 9:** Testing and Feedback Loop

**Step – 10:** Finalization

**PROGRAM**

**HTML**

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Sample Design System</title>
```

```
<style>
    body {
        font-family: 'Arial', sans-serif;
        line-height: 1.6;
        margin: 0;
        padding: 0;
    }
    header {
        background-color: #333;
        color: #fff;
        padding: 1rem;
        text-align: center;
    }
    section {
        margin: 2rem;
    }
    h2 {
        color: #333;
    }
    p {
        color: #555;
    }
    .color-swatch  {
        width:  50px;
        height: 50px;
```

```
            margin: 0 10px 10px 0;
            display: inline-block;
            border: 1px solid #ccc;
        }
    </style>
</head>
<body>
    <header>
        <h1>Sample Design System</h1>
    </header>
    <section>
        <h2>Mood Board</h2>
        <p>Collect inspiration from various sources and present it in a visually appealing way.</p>
        <!-- Insert mood board images here -->
    </section>
    <section>
        <h2>Fonts</h2>
        <p>Choose primary and secondary fonts, define sizes and weights, and document in a style guide.</p>
        <!-- Insert typography style guide here -->
    </section>
    <section>
        <h2>Colors</h2>
        <p>Identify brand colors, establish relationships, and ensure accessibility. Document in a palette.</p>
        <div class="color-swatch" style="background-color: #3498db;"></div>
```

```
        <div class="color-swatch" style="background-color: #2ecc71;"></div>
        <div class="color-swatch" style="background-color: #e74c3c;"></div>
        <!-- Add more color swatches -->
    </section>
    <section>
        <h2>UI Principles</h2>
        <p>Identify user needs, prioritize simplicity and consistency. Document key principles.</p>
        <!-- Insert UI principles documentation here -->
    </section>
    <!-- Add more sections for additional components of your design system -->
</body>
</html>
```

**CSS**

```
body {
    font-family: 'Arial', sans-serif;
    line-height: 1.6;
    margin: 0;
    padding: 0;
}
header {
    background-color: #333;
    color: #fff;
    padding: 1rem;
    text-align: center;
}
```

```
section {
    margin: 2rem;
}
h2 {
    color: #333;
}
p {
    color: #555;
}
.color-swatch  {
    width:  50px;
    height: 50px;
    margin: 0 10px 10px 0;
    display: inline-block;
    border: 1px solid #ccc;
}
```

**JAVASCRIPT**

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Sample Design System</title>
    <style>
        /* Your existing CSS goes here */
```

```
    </style>
</head>
<body>
    <header>
        <h1>Sample Design System</h1>
    </header>
    <button onclick="toggleSection('moodBoard')">Toggle Mood Board</button>
    <button onclick="toggleSection('fonts')">Toggle Fonts</button>
    <button onclick="toggleSection('colors')">Toggle Colors</button>
    <button onclick="toggleSection('uiPrinciples')">Toggle UI Principles</button>
    <section id="moodBoard">
        <!-- Mood Board content goes here -->
    </section>
    <section id="fonts">
        <!-- Fonts content goes here -->
    </section>
    <section id="colors">
        <!-- Colors content goes here -->
    </section>
    <section id="uiPrinciples">
        <!-- UI Principles content goes here -->
    </section>
    <script>
        function toggleSection(sectionId) {
            var section = document.getElementById(sectionId);
```

```
        section.style.display = (section.style.display === 'none' || section.style.display === '') ?
'block' : 'none';

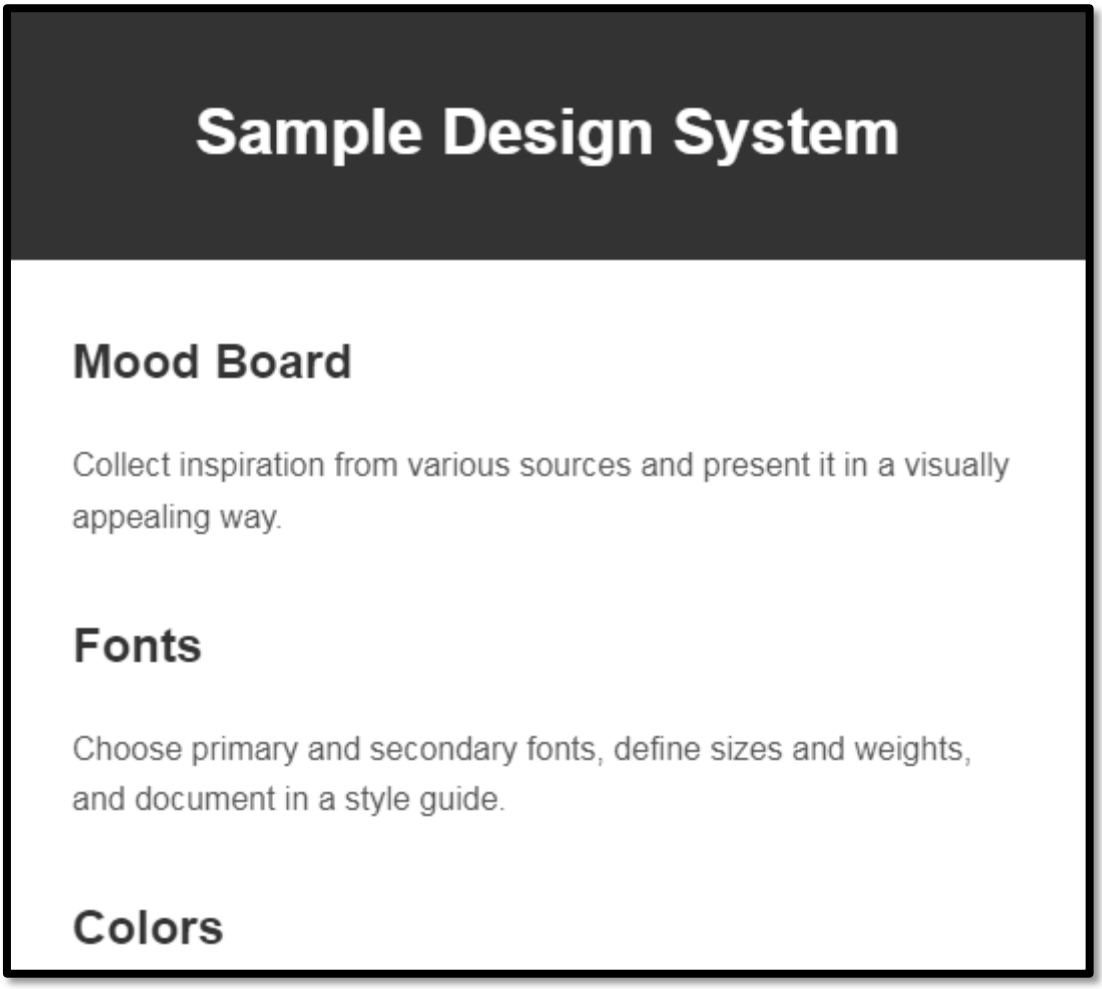
    }

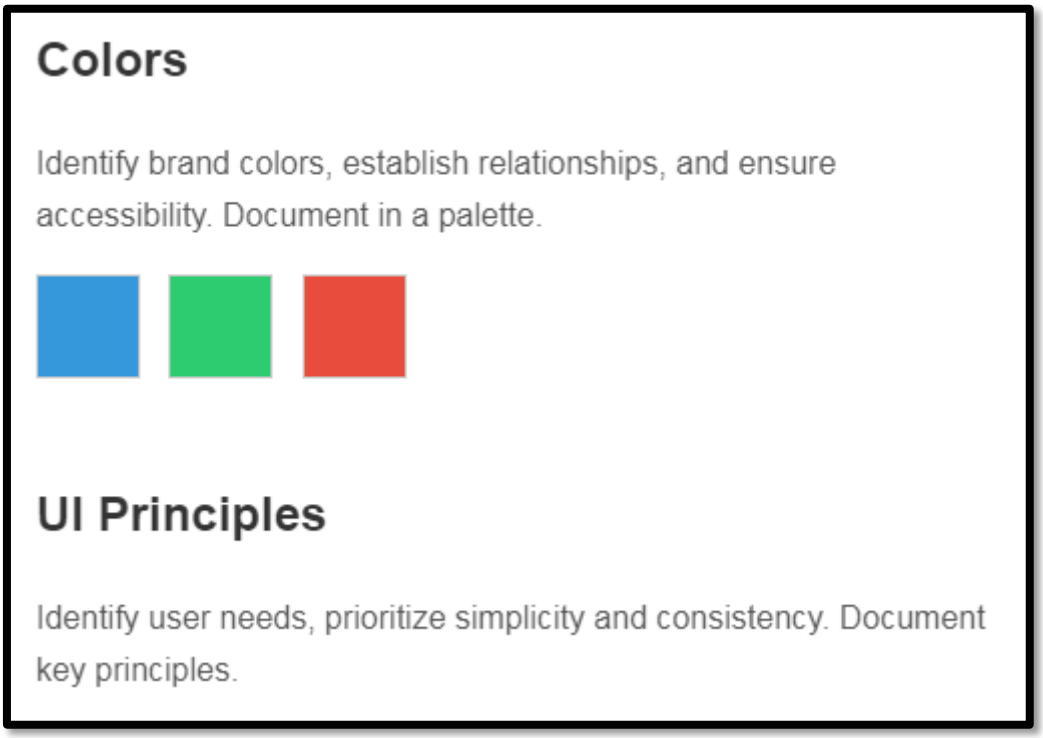
  </script>

</body>

</html>
```

**OUTPUT**






**RESULT**

The sample pattern library results in Consistency, Efficiency, User Experience, Collaboration, Brand Identity, Best Practices.

| EX. NO: 10<br>DATE: | Identify a customer problem to solve |
|---|---|

**AIM**

The primary aim is to enhance user engagement and reduce bounce rates on product pages through a comprehensive UI/UX redesign. This involves addressing identified pain points, optimizing performance, and implementing design improvements to create a more compelling and user-friendly experience. The goal is to positively impact user satisfaction, increase conversion rates, and contribute to the overall success of the digital product.

**PROCEDURE**

**Step – 1:** User Research

**Step – 2:** Identify Common Issues

**Step – 3:** Competitor Analysis

**Step – 4:** Prioritize Issues

**Step – 5:** Optimize Page Load Times

**Step – 6:** Simplify Layout

**Step – 7:** Enhance Visual Hierarchy

**Step – 8:** Mobile Responsiveness

**Step – 9:** A/B Testing

**Step – 10:** Iterative Improvement

**PROGRAM**

**HTML**

```
<!DOCTYPE html>

<html lang="en">

<head>

    <meta charset="UTF-8">

    <meta name="viewport" content="width=device-width, initial-scale=1.0">
```

```
    <link rel="stylesheet" href="styles.css">
    <script defer src="script.js"></script>
    <title>Your Product Page</title>
</head>
<body>
    <header>
        <h1>Product Name</h1>
    </header>
    <main class="product-container">
        <section class="product-details">
            <img data-src="https://picsum.photos/800/600" alt="Product Image">
            <h2>Product Description</h2>
            <p>Lorem ipsum dolor sit amet, consectetur adipiscing elit. Sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.</p>
            <button>Add to Cart</button>
        </section>
        <section class="related-products">
            <h2>Related Products</h2>
            <!-- Add related product links or images here -->
        </section>
    </main>
    <footer>
        <p>&copy; 2023 Your Company Name. All rights reserved.</p>
    </footer>
</body>
```

```
</html>
```

**CSS**

```
body {
    font-family: 'Arial', sans-serif;
    line-height: 1.6;
    background-color: #f8f8f8;
    color: #333;
    margin: 0;
}
header, footer {
    background-color: #4CAF50;
    color: #fff;
    text-align: center;
    padding: 1em;
}
.product-container {
    max-width: 800px;
    margin: 0 auto;
    background-color: #fff;
    padding: 1em;
    box-shadow: 0 0 10px rgba(0, 0, 0, 0.1);
}
.product-details img {
    width: 100%;
    height: auto;
```

```
}
.product-details button {
    background-color: #4CAF50;
    color: #fff;
    padding: 10px;
    border: none;
    cursor: pointer;
}
.product-details button:hover {
    background-color: #45a049;
}
.related-products {
    margin-top: 2em;
}
footer {
    position: fixed;
    bottom: 0;
    width: 100%;
}
```

**JAVASCRIPT**

```
document.addEventListener("DOMContentLoaded", function() {
    const images = document.querySelectorAll("img[data-src]");
    const lazyLoad = target => {
        const io = new IntersectionObserver((entries, observer) => {
            entries.forEach(entry => {
```

```
            if (entry.isIntersecting) {
                const img = entry.target;
                img.src = img.dataset.src;
                observer.disconnect();
            }
        });
    });
    io.observe(target);
};
images.forEach(lazyLoad);
});
```

**OUTPUT**





## Product Description

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.

Add to Cart

© 2023 Your Company Name. All rights reserved.

**RESULT**

The implementation of the redesigned product page with a focus on addressing high bounce rates yields positive outcomes Reduced Bounce Rates, Increased Engagement, Improved Conversion Rates, Positive User Feedback, Enhanced Brand Perception, and Data-Driven Decision-Making.

| EX. NO: 11<br>DATE: | **Conduct end-to-end user research - User research, creating personas, Ideation process (User stories, Scenarios), Flow diagrams, Flow Mapping** |
|---|---|

**AIM**

The aim is to create a seamless and enjoyable user experience by combining user insights, persona-driven design, ideation, and a clear visualization of user flows throughout the design and development process.

**PROCEDURE**

**Step – 1:** User Research

**Step – 2:** Creating Personas

**Step – 3:** Ideation Process (User Stories, Scenarios)

**Step – 4:** Flow Diagrams

**Step – 5:** Flow Mapping

**Step – 6:** Result

**PROGRAM**

**PYTHON**

```
# Import necessary libraries

import pandas as pd

# Step 1: Define Objectives and Develop Research Tools

research_objectives = ["Understand user preferences", "Identify pain points"]

survey_questions = [

    "What features do you find most useful?",

    "What challenges do you face with the current product?",

]

# Step 4: Conduct Research

def conduct_survey(objectives, questions):
```

```
    print("Conducting survey...")
    survey_responses = {}
    for objective in objectives:
        print(f"\nObjective: {objective}")
        responses = {}
        for question in questions:
            response = input(f"{question} ")
            responses[question] = response
        survey_responses[objective] = responses
    return survey_responses
# Step 5: Analyze Data
def analyze_data(survey_responses):
    print("\nAnalyzing survey data...")
    df = pd.DataFrame(survey_responses)
    print(df)
# Step 6: Create Personas
def create_personas():
    # Implementation of persona creation would depend on data analysis and business rules
    print("Creating personas...")
# Step 3: Ideation Process
def generate_user_stories():
    print("Generating user stories...")
    # Your implementation for generating user stories goes here
# Step 4: Flow Diagrams
def map_high_level_flows():
```

```
    print("Mapping high-level user flows...")
    # Your implementation for creating flow diagrams goes here
# Step 5: Flow Mapping
def create_detailed_flow_maps():
    print("Creating detailed flow maps...")
    # Your implementation for creating detailed flow maps goes here
# Step 6: Result
def visualize_results():
    print("Visualizing results...")
    # Your implementation for visualizing results goes here
# Main execution
survey_responses = conduct_survey(research_objectives, survey_questions)
analyze_data(survey_responses)
create_personas()
generate_user_stories()
map_high_level_flows()
create_detailed_flow_maps()
visualize_results()
```

**OUTPUT**

```
Conducting survey...

Objective: Understand user preferences
What features do you find most useful? Easy navigation
What challenges do you face with the current product? Slow loading times

Objective: Identify pain points
What features do you find most useful? Search functionality
What challenges do you face with the current product? Confusing layout

Analyzing survey data...
  Understand user preferences   Identify pain points
0               Easy navigation  Search functionality
1           Slow loading times        Confusing layout

Creating personas...
Generating user stories...
Mapping high-level user flows...
Creating detailed flow maps...
Visualizing results...
```

**RESULT**

It is a well-informed, user-centered design that aligns with both user expectations and business goals. The end product is more likely to be embraced by users, leading to increased user satisfaction and the potential for business success. The iterative nature of the process also positions the design for continuous improvement based on evolving user needs and market dynamics.

| EX. NO: 12<br>DATE: | **Sketch, design with popular tool and build a prototype and perform usability testing and identify improvements** |
|---|---|

## AIM

Create an optimal user experience (UX) by refining and validating design choices through sketching, high-fidelity designs, and prototypes. Conduct usability testing to identify and implement improvements, ensuring stakeholder alignment and preparing for efficient development. Deliver a user-validated design that exceeds user expectations.

## PROCEDURE

**Step – 1:** Sketching

**Step – 2:** Design with Popular Tools

**Step – 3:** Build a Prototype

**Step – 4:** Usability Testing

**Step – 5:** Identify Improvements

**Step – 6:** Iterative Design

**Step – 7:** Stakeholder Alignment

**Step – 8:** Prepare for Development

**Step – 9:** Deliver User-Validated Design

## PROGRAM

## HTML, CSS AND JAVASCRIPT

```
<!DOCTYPE html>

<html lang="en">

<head>

  <meta charset="UTF-8">

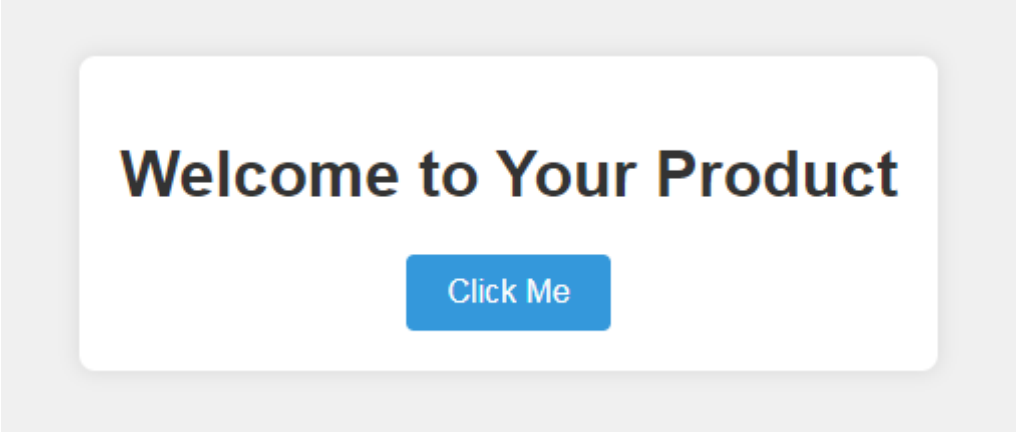
  <meta name="viewport" content="width=device-width, initial-scale=1.0">

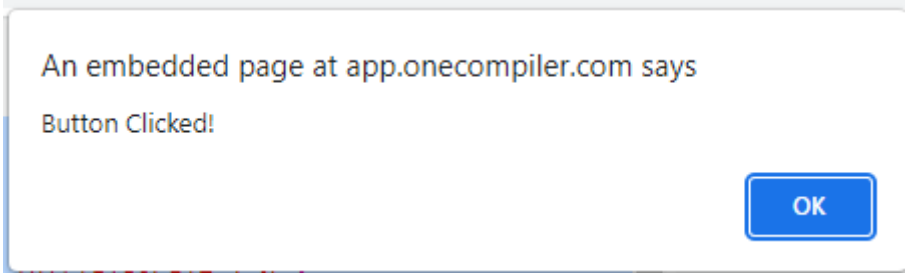
  <title>UX Design Process</title>
```

```
<style>
  body {
    font-family: 'Arial', sans-serif;
    margin: 0;
    padding: 0;
    background-color: #f0f0f0;
    display: flex;
    justify-content: center;
    align-items: center;
    height: 100vh;
  }
  #app {
    background-color: #fff;
    border-radius: 8px;
    padding: 20px;
    box-shadow: 0 0 10px rgba(0, 0, 0, 0.1);
    text-align: center;
  }
  h1 {
    color: #333;
  }
  button {
    background-color: #3498db;
    color: #fff;
    padding: 10px 20px;
```

```
      font-size: 16px;
      border: none;
      border-radius: 4px;
      cursor: pointer;
    }
    button:hover {
      background-color: #2980b9;
    }
  </style>
</head>
<body>
  <div id="app">
    <h1>Welcome to Your Product</h1>
    <button onclick="simulateButtonClick()">Click Me</button>
  </div>
  <script>
    function simulateButtonClick() {
      alert('Button Clicked!');
    }
  </script>
</body>
</html>
```

**OUTPUT**




**RESULT**

It is a well-informed, refined, and user-validated UI/UX design, poised for successful implementation. The integration of user feedback and iterative design ensures that the final product aligns closely with user needs and expectations, ultimately contributing to a positive user experience and the success of the project.